V1.3 RUN: nominal3.3V header supply. NO USB / external5V while seated.
POWER_GOOD holds CHIP_PU low until both monitored rails stabilize.
GPIO4 READY must remain0 until all MCU ADC-side output values are initialized.

F21: probe access only; test-pad footprints/layout pending

F09: 10k pulldowns bias START, RESET and PWDN low while MCU is high-Z.
CS defaults / complete power sequencing remain unresolved.

V1.3 POWER IMPLEMENTED IN SCHEMATIC – BENCH QUALIFICATION REQUIRED
MAIN 3.3V for stock DevKit; independent4.0V PRE –> quiet2.98V AVDD+DVDD.
Battery–only while seated: both USB sockets physically blocked; unplug pack/unseat U3 for USB service.
GPIO4 READY + power monitors isolate ADC controls/DOUT during reset or rail faults.
F06/F18: initialize U1/CLK_EN then U2; enable/settle2.4V references before common START.
F07/F08: protected body bias and wearer fault–current protection remain open.

V1.3 F12/F14 – PROCUREMENT REQUIREMENTS (not qualified installed parts)
R1–R32: 2k unchanged; 0.1%, >=0.1W, derating/pulse review required.
All 32 input 620pF caps: C0G/NP0, 1%; electrode impedance/CMRR still unverified.
C25/C51 VCAP3: 1uF unchanged, >=10V; qualify capacitance at ~5.2V DC.
C55/BP obsolete regulator removed; qualify new SET leakage/microphonics.
New MPNs are schematic selections; effective C, footprints/layout and physical tests remain required.

V1.3 F09: receiver–side DIN/SCLK/START/RESET/PWDN pulldowns retained; CS pulldowns added.
Old U4/U5/R38 removed after candidate rail/control validation. New EN has defined UVLO and inhibit link.
No mounted USB mode: mechanical connector exclusion is required, not electrical USB hotplug protection.

BATTERY ENTRY – specified external protected1S pack; unplug pack for OFF

J17: pin1 POSITIVE, pin2 RETURN. Verify cable polarity; no universal Li–Po connector polarity.
V1.3 POWER: protected 1S pack / bench3.0–4.2V. NO USB while seated. Unplug pack for OFF/service.
F1/Q1 add secondary input and polarity protection; Q1 does not block reverse current. No USB/charging while worn.

JP1 normally OPEN. Short for service inhibit only; disconnect battery and unseat DevKit before USB service.

U6:3.3V main; U7:independent4.0V pre; U8:quiet2.98V AVDD+DVDD. Outputs never paralleled.

22k/10k MAIN EN cutoff is provisional: about3.10–3.31V falling at switched/protected input; pack qualification required.

MAIN_PG sequences preconverter and LDO; separate supervisors/READY gate must isolate every MCU/ADC path on rail/reset fault.

PCB/BOM release gate: exact TPS DLA0010A+inductor footprints, MLCC effective–C/ESR/leakage, inrush/UVLO/rebound/noise/thermal tests.

Local ADC bypass parts: place C112/C114 at U1 AVDD1/AVSS1 and C113/C115 at U2.
C110/C111 add100nF at VCAP3 in parallel with retained1uF. C57 is polarized220uF MAIN bulk.
Physical placement, effective capacitance and startup/noise remain PCB/bench release gates.

Version 1.3 – power implementation; bench qualification pending
McGill AI Lab – Researcher: Karen Chen Lai
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File: open–semg–16–v1.3.kicad_sch
Title: sEMG Wirstband
Size: A0 | Date: | **Rev:**
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